Thirsty Engines: Why Your 1974 Car Hates Your Wallet

Jozef Rivest

University of Michigan

Motivation

Humanity's oldest question: *why am I at the gas station again?* Your uncle with the 1973 Cadillac has opinions. This poster has data.

Puzzle

The Toyota Corolla gets **33.9** miles per gallon. The Cadillac Fleetwood gets **10.4**. What is the Cadillac *doing* with all that gas?

A bit of theory

Two mechanisms explain engine thirst:

1. **Many Mouths:** each cylinder is a tiny mouth that drinks gasoline
2. **Enthusiasm:** horsepower is *how hard* those mouths drink

H1: More cylinders mean fewer miles per gallon.

H2: Holding cylinders constant, more horsepower means fewer miles per gallon.

Background

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Related work

- Lorem ipsum dolor sit amet, consectetur adipiscing elit
- Sed do eiusmod tempor incididunt ut labore et dolore
- Ut enim ad minim veniam, quis nostrud exercitation
- Duis aute irure dolor in reprehenderit in voluptate

Curabitur pretium tincidunt lacus. Nulla gravida orci a odio. Nullam varius, turpis et commodo pharetra, est eros bibendum elit, nec luctus magna felis sollicitudin mauris. Integer in mauris eu nibh euismod gravida.

Scope conditions

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Design

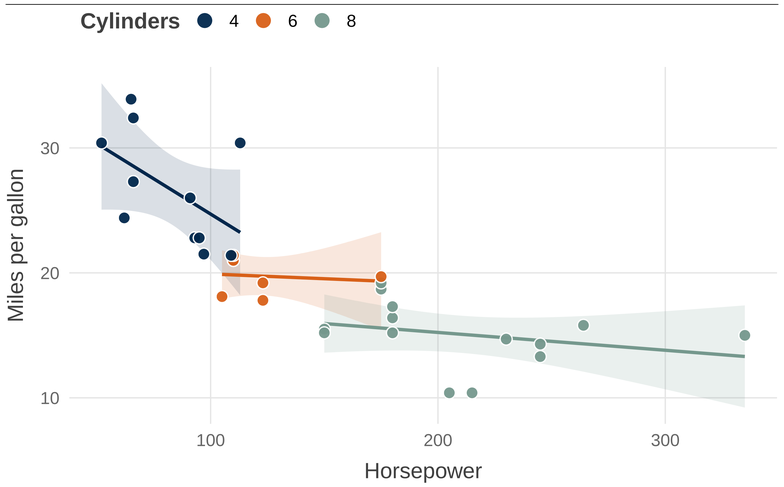
I estimate the following model with OLS, because I know how to do it:

$$\text{MPG}_i = \beta_0 + \beta_1 \text{Cyl}_i + \beta_2 \text{HP}_i + \varepsilon_i$$

Data

- `mtcars`: 32 cars road-tested by *Motor Trend* in 1974
- Outcome: miles per gallon (higher = smugger)
- Predictors: cylinders (mouths) and horsepower (enthusiasm)

Results 1



Robustness

Curabitur pretium tincidunt lacus. Nulla gravida orci a odio. Nullam varius, turpis et commodo pharetra, est eros bibendum elit, nec luctus magna felis sollicitudin mauris. Integer in mauris eu nibh euismod gravida.

1. Lorem ipsum dolor sit amet, consectetur adipiscing elit
2. Sed do eiusmod tempor incididunt ut labore et dolore magna
3. Ut enim ad minim veniam, quis nostrud exercitation ullamco

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Heterogeneity

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Results 2

H1 confirmed: an 8-cylinder car loses about 8.5 mpg relative to a 4-cylinder car.

	Model 1
Constant	28.650***
	(1.588)
6 cylinders	-5.968**
	(1.639)
8 cylinders	-8.521**
	(2.326)
Horsepower	-0.024
	(0.015)
Num.Obs.	32
R2	0.754

Conclusion

- More cylinders, fewer miles per gallon
- **H2 runs out of gas:** horsepower is guilty in spirit, but $p \approx 0.13$
- Future work: data from after 1974, and asking who *needs* a V8 for groceries

Vroom?

Next steps

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Curabitur pretium tincidunt lacus. Nulla gravida orci a odio. Nullam varius, turpis et commodo pharetra, est eros bibendum elit, nec luctus magna felis sollicitudin mauris. Integer in mauris eu nibh euismod gravida.

Acknowledgements

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

References

- Lorem, I. (2025). *Dolor Sit Amet*. Consectetur Press.
- Ipsum, L., & Dolor, S. (2024). Adipiscing elit sed do eiusmod. *Journal of Tempor Studies*, 12(3), 45–67.
- Amet, C. (2023). Ut labore et dolore magna aliqua. *Review of Veniam*, 8(1), 1–22.